Engineering Project Management & Finance

UESTC3031 & UESTCHN3012

(2024 – 2025 Semester 2)

Assignment title:

Enhancing Engineering Management Learning: Soft Skills Development

(Assessment: Report)

The numbers in square brackets in the right-hand margin indicate the marks allotted to each part of the question against which the mark is shown. These marks are for guidance only. The total mark in this report is calculated based on 100 and then will be scaled considering the second assignment from other course contents.

Purpose of the Assignment

The 3rd-year course, Engineering Project Management and Finance (EPMF), is designed to equip students with the core principles of Engineering Management (EM). Through the course lectures, students will gain a comprehensive understanding of both project management and design for manufacturing (DFM). This foundation will prepare them with the necessary skills for effective decision-making and problem-solving in their future engineering careers.While the EPMF course provides lectures and assignments that introduce and reinforce key EM skills, students require further opportunities to apply their knowledge in practical settings. Without such opportunities after completing EM lectures, students may struggle to fully internalize and refine their skills. The current coursework aims to explore these possibilities and propose measurable ways to provide additional hands-on practice, strengthening their learning outcomes.

Date Issued to Students:

- ***Chengdu Campus:*** *14th March 2025 (Friday),* ***Hainan Campus:*** *27th March 2025 (Thursday)*

Deadline / Submission Date:

- ***Chengdu Campus:*** *28th March 2025 (Friday, 11:59 PM CST),* ***Hainan Campus:*** *10th April 2025 (Thursday, 11:59 PM CST)*

Submission Instructions:

- You must submit in **ONE PDF document** (not a zipped file, not two times, not in a Word document) as the report of this assignment (only 1 submission in a system for each student is allowable).
- You need to complete the assignment report in the provided **Answering Booklet** file maintaining the relevant format. You can find the Answering Booklet alongside this submitted assignment. Each solution corresponding to each question must not exceed 250 words.
- You must prepare this assessment **individually (not in a group/team)**. In the file name, indicate your name and ID in the following format **StudentName_GUID_UESTCID.pdf.** Inside the document, you need to provide the **student's name, GUID, and UESTCID** so that we know who submitted the report.
- You must submit a **plagiarism declaration** and your submission will go through plagiarism check.
- **No late submissions will be allowed in Moodle** so if you don't submit you will get a zero for this complete assessment.
- This is a **SUMMATIVE assessment** (i.e., marked assessment). The marking criteria are provided.

<< Dr. Amir >>

Assessment Report [100 Marks] **(Note: The total mark for this report is based on 100 and will be scaled regarding the second assignment)**.

Title: Enhancing Engineering Management Learning: Soft Skills Development

Objective: This coursework utilizes scenarios to simulate practical cases, assisting students in applying Engineering Management skills. It enhances their understanding of project management and design for manufacturing while improving their decision-making abilities.

Open-Ended Questions:

1. Question 1 [10/100]

👉 **Scenario:** Your company is designing a new smart home device. You have two material options for the outer casing:

- **Option A:** Recyclable biodegradable plastic, but 30% more expensive.
- **Option B:** Standard ABS plastic, lower cost but not environmentally friendly.

  As a DFM engineer, how do you approach this decision? What factors do you consider?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Problem Solving, Decision-Making, Clarity of Response (Presentation Skills)

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2. Question 2 [10/100]

👉 **Scenario:** Your team is producing high-frequency power converters for an EV (Electric Vehicle) charging station. A batch fails thermal stress tests, causing overheating at high loads. The Quality team suggests more testing, while the Manufacturing team wants to modify the heat sink design.

- As the lead engineer, how do you decide?
- How do you communicate with both teams?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Problem Solving, Risk Management, Communication & Stakeholder Engagement & Leadership, Clarity of Response (Presentation Skills)

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3. Question 3 [10/100]

👉 **Scenario:** Your company is designing a high-efficiency PCB (Printed Circuit Board) for sensors used in connected devices, such as smart home appliances or industrial systems. A supplier informs you of ±10% variability in copper trace thickness.

- How would you apply the Taguchi robust design principles to ensure the PCB performs consistently despite this variability?
- How would you collaborate with the supplier to minimize the impact of this variability on the final product?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Teamwork, Problem Solving, Clarity of Response (Presentation Skills)

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4. Question 4 [10/100]

👉 **Scenario:** Your company produces high-precision components for medical devices. The current defect rate is higher than expected, and the process seems to have variability. You are tasked with improving the process to ensure fewer defects and consistent quality.

- How would you assess the performance of the process to identify areas for improvement?

- What steps would you take to reduce defects and increase the process's ability to consistently meet quality standards?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Teamwork, Problem Solving, Presentation Skills (Clarity of Response)

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5. Question 5 [10/100]

👉 **Scenario:** Your company is manufacturing resistors, and you are noticing some inconsistencies in the output. You have been tasked with implementing Statistical Process Control (SPC) to monitor and control the production process.

- How would you use control charts to identify potential issues in the production process?
- What actions would you take if you noticed the process is out of control based on the control charts?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Teamwork, Problem Solving, Presentation Skills (Clarity of Response)

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6. Question 6 [10/100]

👉 **Scenario:** Your team is working on a project to develop a smart metering system. A stakeholder requests a last-minute addition of AI-based energy prediction features, requiring an extra two months of work.

- How do you handle this request?
- How do you communicate the impact?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Stakeholder Negotiation, Risk Management, Presentation Skills (Clarity of Response)

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7. Question 7 [10/100]

👉 **Scenario:** A key embedded systems engineer on your drone navigation project resigns unexpectedly. Your deadline is two months away, and hiring a replacement takes six weeks.

- How do you reallocate resources?
- How do you ensure project deadlines are met?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Stakeholder Negotiation, Risk Management, Presentation Skills (Clarity of Response)

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8. Question 8 [10/100]

👉 **Scenario:** Your company is developing a 5G signal amplifier. The project is over budget by 15%, and management asks for cost reductions without reducing performance.

- How do you identify cost-cutting areas?
- How do you justify your decisions to management?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Stakeholder Negotiation, Risk Management, Presentation Skills (Clarity of Response)

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9. Question 9 [10/100]

👉 **Scenario:** Your team is working on a high-priority software development project with tight deadlines. The project schedule is falling behind due to resource constraints, particularly a shortage of skilled developers. Management asks you to expedite the project delivery without compromising quality.

- How do you assess the impact of resource constraints on the project schedule?

- What steps would you take to manage both the schedule and resources efficiently?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Project Management, Critical Thinking, Communication & Stakeholder Negotiation, Risk Management, Presentation Skills (Clarity of Response)

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10. Question 10 [10/100]

👉 **Scenario:** During your project, two key stakeholders—one from the client side and one from the development team, have a significant disagreement. The client insists on additional features being included, which will extend the project timeline and increase costs, while the development team is concerned about the feasibility and potential impact on existing functionality. As the project leader, you must resolve this conflict without disrupting the project's progress.

- How do you manage this conflict between the two stakeholders while maintaining a positive team dynamic and client relationship?
- How do you ensure that both parties' concerns are addressed and that the project remains on track?

💡 ***Soft Skills Assessed:*** Leadership & Teamwork, Conflict Management, Communication & Stakeholder Negotiation, Communication & Emotional Intelligence, Presentation Skills

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Deliverables:

Submit one PDF report following the provided Answering Booklet format. The report must include detailed responses to each question while maintaining the required structure. Submission must be in a single PDF document—Word documents, zipped files, or multiple submissions will not be accepted. Each student is allowed only one submission in the system. The assessment must be completed individually, not in a group. The file name must follow the format StudentName_GUID_UESTCID.pdf, and the document must include the student's name, GUID, and UESTCID for identification.

Grading criteria:

The submitted report will be evaluated based on the quality, accuracy, and depth of responses. Each open-ended question will be assessed separately using specific criteria aligned with the soft skills outlined in the question, ensuring a comprehensive evaluation of technical accuracy, critical thinking, communication, and other relevant competencies.

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